Mystery benchmarks

SCC 2024




Recap: Benchmarking

- Testing the system as a whole, or a specific subsystem
  - CPUs and GPUs, memory or I/O
- Does some sort of calculations, but might not be anything scientifically relevant
  - A large number of randomized matrix operations is a common approach
- Nodes connected via an interconnect that transfers data

Testing processing power

- Are you testing double precision or lower precision performance?
  - e.g. HPL vs MLPerf/HPL-MxP
- In computation-bound programs, the bottleneck is mainly the CPU and GPU execution speed
  - E.g. HPL
  - Computational intensity is high
- In memory-bound programs, memory bandwidth becomes the bottleneck
  - Sparse matrix-vector multiplication (e.g. HPCG)
  - Computational intensity is low

Testing random-access memory

- Benchmarking *sustainable* memory bandwidth
- STREAM has been the industry standard
- Tests a variety of aspects:
  - "Copy" measures transfer rates in the absence of arithmetic.
  - "Scale" adds a simple arithmetic operation.
  - "Sum" adds a third operand to allow multiple load/store ports on vector machines to be tested.
  - "Triad" allows chained/overlapped/fused multiply/add operations.
- "BabelStream" implements this for (AMD) GPUs
  - https://www.amd.com/en/developer/resources/infinity-hub/babelstream.html

BabelStream

Testing the I/O

- Which aspect of the I/O are we testing?
- IO500 has been gaining popularity
  - Tests multiple aspects of the disk
  - Randomized search, search following a pattern, creating and removing large number of files, etc.
  - Combined score based on overall performance
- Essentially a benchmark suite for various tests
  - **IOEasy**: Applications with well optimized I/O patterns
  - **IOHard**: Applications that require a random workload
  - **MDEasy**: Metadata/small objects
  - **MDHard**: Small files (3901 bytes) in a shared directory
  - **Find**: Finding relevant objects based on patterns

IO500

What else?

- OSU micro-benchmarks for interconnect bandwidth testing
  - Supports infiniband, slingshot and omnipath
  - Good tools for testing your system interconnect
- HPC Challenge benchmark
  - Combines a variety of tests to test the system thoroughly
  - Should be tested
- Benchmarking scientific software
  - Many software offer their own "benchmarking" inputs for testing how their application runs on a given system
  - Gromacs, OpenFoam, Gpaw...

My guess for the mystery benchmark

1. HPCG
2. STREAM
3. IO500
4. HPC Challenge
5. HPL-MxP